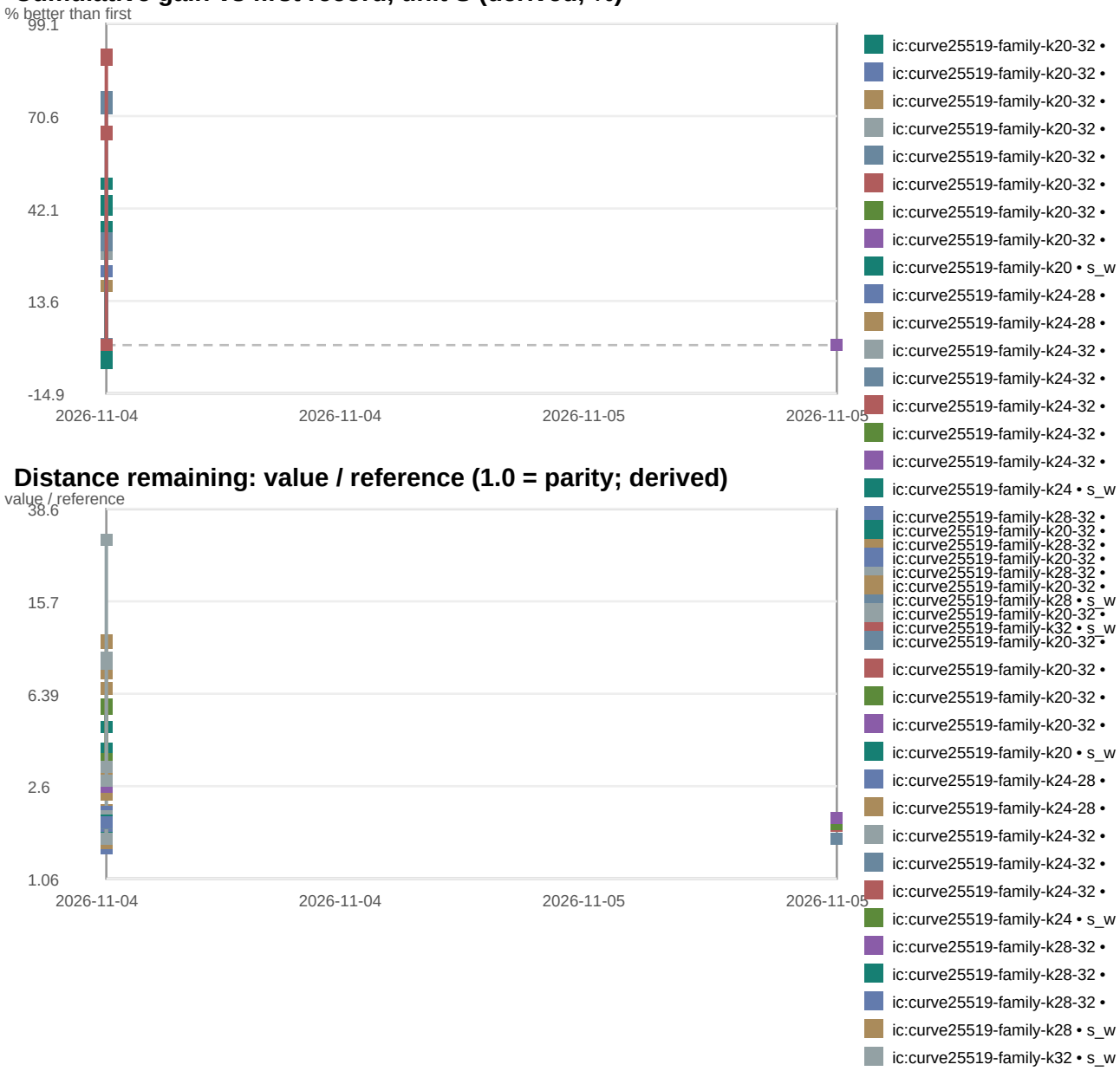
# Benchmark progress: crypto

From bench.record/v1 records in Git; raw data in S3 by SHA-256. Derived numbers are labelled derived.

Records: open 0, measured 39, invalid 0, censored 3. Headline unit S: group operations / sqrt(n); the ECDLP cost unit, every phase priced.

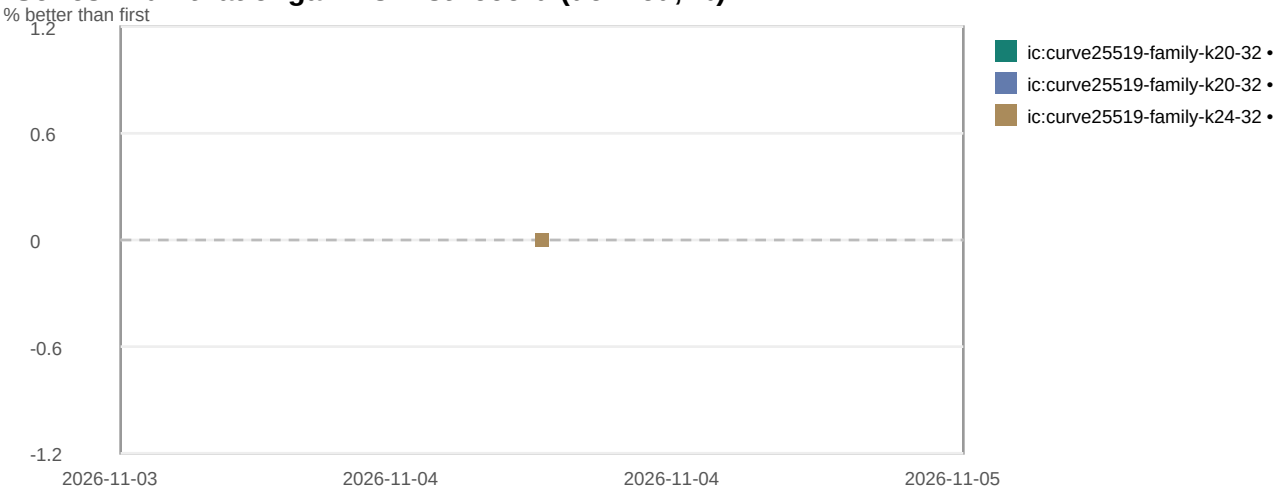
## Cumulative gain vs first record, unit S (derived, %)




## Distance remaining: value / reference (1.0 = parity; derived)

## Series in unit count: gain vs first record (derived, %)

## Series in unit ratio: gain vs first record (derived, %)



## Series summary

| problem • metric | unit | first | latest | gain vs first | value/ref | pts |
|---|---|---|---|---|---|---|
| archive:icb-large-prime-n53-s1-20261029 • report | count | 240 | 240 | ? | ? | 1 |
| archive:icb-large-prime-n53-s2-20261029 • report | count | 240 | 240 | ? | ? | 1 |
| archive:icb-large-prime-n61-s1-20261029 • report | count | 72 | 72 | ? | ? | 1 |
| archive:icb-large-prime-n61-s2-20261029 • report | count | 60 | 60 | ? | ? | 1 |
| ic:curve25519-family-k20-32 • ratio_w0_k24 | ratio | 0.9375 | 0.9375 | 0% | ? | 1 |
| ic:curve25519-family-k20-32 • ratio_w0_k32 | ratio | 0.8802 | 0.8802 | 0% | ? | 1 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k20 | S | 2.5746 | 2.5746 | 0% | 1.7601 | 1 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k24 | S | 2.1638 | 1.6735 | 22.6592% | 1.429 | 2 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k28 | S | 1.9211 | 1.5717 | 18.1849% | 1.4943 | 2 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k32 | S | 2.11 | 1.5152 | 28.1881% | 1.5539 | 2 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 2.601 | 2.601 | 0% | 1.5669 | 1 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 2.111 | 2.111 | 0% | 1.7695 | 1 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 1.869 | 1.869 | 0% | 1.7971 | 1 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 1.93 | 1.93 | 0% | 1.9052 | 1 |
| ic:curve25519-family-k20 • s_w0_incumbent | S | 8.3478 | 4.8598 | 41.7827% | 3.2356 | 5 |
| ic:curve25519-family-k24-28 • s_w0_incumbent_k24 | S | 2.5842 | 2.5842 | 0% | 2.0304 | 1 |
| ic:curve25519-family-k24-28 • s_w0_incumbent_k28 | S | 2.3732 | 2.3732 | 0% | 2.3865 | 1 |
| ic:curve25519-family-k24-32 • best_train_s_w0_k2 | S | 2.713 | 2.713 | 0% | ? | 1 |
| ic:curve25519-family-k24-32 • best_train_s_w0_k3 | S | 2.768 | 2.768 | 0% | ? | 1 |
| ic:curve25519-family-k24-32 • ratio_w0_k32 | ratio | 0.9106 | 0.9106 | 0% | ? | 1 |
| ic:curve25519-family-k24-32 • s_w0_seedrun_k24 | S | 2.1055 | 2.1055 | 0% | 1.942 | 1 |
| ic:curve25519-family-k24-32 • s_w0_seedrun_k28 | S | 1.8835 | 1.8835 | 0% | 1.8154 | 1 |
| ic:curve25519-family-k24-32 • s_w0_seedrun_k32 | S | 1.8423 | 1.8423 | 0% | 1.8232 | 1 |
| ic:curve25519-family-k24 • s_w0_incumbent | S | 6.607 | 3.3182 | 49.7782% | 2.7163 | 5 |
| ic:curve25519-family-k28-32 • s_w0_incumbent | S | 2.5798 | 2.5798 | 0% | 2.5891 | 1 |
| ic:curve25519-family-k28-32 • s_w0_seed_mean_k28 | S | 1.969 | 1.969 | 0% | 1.8593 | 1 |
| ic:curve25519-family-k28-32 • s_w0_seed_mean_k32 | S | 1.781 | 1.781 | 0% | 1.8229 | 1 |
| ic:curve25519-family-k28 • s_w0_incumbent | S | 10.9395 | 2.576 | 76.452% | 2.7572 | 7 |
| ic:curve25519-family-k32 • s_w0_incumbent | S | 24.1377 | 2.519 | 89.5641% | 2.7453 | 5 |

Every value is quoted from the record it names, in that record’s unit. Percentages and ratios under ’derived’ are computed here from those values and are not measurements. Series in different units are never compared.